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Computer Science - 9618

■ 35 Questions

Topics Covered :

✓ 3.1 - Computers and Their Components

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Question 1 - (9618_s25_11_4)

Subsection Topics: (a) : 7.1 - Computing Ethics and Ownership | (b) : 3.1 - Computers and Their Components

A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer’s account.

(a) The new system uses digital cameras and Artificial Intelligence (AI) to identify the customers.

Explain how the new system uses AI to identify each customer.

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..... [4]

(b) The new system uses sensors to identify the items taken from a shelf.

Identify **one** type of sensor that can be used in this new system.

State how the sensor can be used to identify the items taken from a shelf.

Sensor

Use

.....

[2]

Question 2 - (9618_s25_11_6)

Subsection Topics: (a) : 3.1 - Computers and Their Components, 4.1 - CPU Architecture | (b) : 5.1 - Operating Systems

A programmer is buying a new computer.

(a) The programmer is considering the following two computers:

Computer 1	Quad-core 2.2 GHz processor 16-bit architecture 1 GB Random Access Memory (RAM) 500 GB magnetic hard disk
Computer 2	Dual-core 3.8 GHz processor 32-bit architecture 2 GB RAM 500 GB solid state drive (SSD)

(i) Computer 1 has a magnetic hard disk.

Complete the description of the principal operation of a magnetic hard disk by writing the missing words.

The magnetic hard disk has one or more that can be magnetised. These are mounted on a and rotate at high speed.

A is moved across the surface on an arm.

When data is read, the changes in the produce a change in the electric current.

[4]

(ii) The two computers have different amounts of RAM.

Explain how different amounts of RAM affect the performance of a computer.

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..... [3]



- [2]

- [5]

Question 3 - (9618_w25_11_8)

Subsection Topics: (a) : 2.1 - Networks Including the Internet | (b) : 3.1 - Computers and Their Components, 5.1 - Operating Systems

A restaurant accesses software and stores data using cloud computing.

(a) (i) Give **two** benefits of storing data using cloud computing.

1

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2

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[2]

(ii) Give **two** drawbacks of using cloud computing.

1

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2

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[2]

(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(i) Explain how a touchscreen converts the point of touch to a selection from the menu.

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[4]

(ii) The image files that are displayed on the touchscreen have been corrupted.

Identify **two** examples of utility software that could be used to recover the images.

1

2

[2]

Question 4 - (9618_w25_11_11)

Subsection Topics: (11) : 3.1 - Computers and Their Components

An automated system opens doors when a person is detected within 2 metres. The system closes the doors when there is no longer a person within 2 metres.

Identify whether the automated system is an example of a monitoring system or a control system.

Justify your choice.

Monitoring or Control

Justification

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..... [3]

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Question 5 - (9618_w25_12_8)

Subsection Topics: (a) : 2.1 - Networks Including the Internet | (b) : 3.1 - Computers and Their Components, 5.1 - Operating Systems

A restaurant accesses software and stores data using cloud computing.

(a) (i) Give **two** benefits of storing data using cloud computing.

1

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2

.....

[2]

(ii) Give **two** drawbacks of using cloud computing.

1

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2

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[2]

(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(i) Explain how a touchscreen converts the point of touch to a selection from the menu.

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[4]

(ii) The image files that are displayed on the touchscreen have been corrupted.

Identify **two** examples of utility software that could be used to recover the images.

1

2

[2]

Question 6 - (9618_w25_12_11)

Subsection Topics: (11) : 3.1 - Computers and Their Components

An automated system opens doors when a person is detected within 2 metres. The system closes the doors when there is no longer a person within 2 metres.

Identify whether the automated system is an example of a monitoring system or a control system.

Justify your choice.

Monitoring or Control

Justification

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[3]

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Question 7 - (9618_w25_13_7)

Subsection Topics: (a) : 7.1 - Computing Ethics and Ownership | (b) : 1.2 - Multimedia | (c) : 6.2 - Data Integrity | (d) : 3.1 - Computers and Their Components | (e) : 6.2 - Data Integrity

A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(a) Describe **one** ethical impact of this use of AI in the classroom.

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..... [2]

(b) A video is made of many bitmap images called frames. 30 frames are recorded every second.

Each frame is 4000 pixels wide by 3000 pixels high. The video records using 16-bit colour depth.

Calculate an estimate for the file size for **one** second of the video in gigabytes.

Show your working.

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File size gigabytes [2]

(c) Identify **and** describe **one** method of data verification that can be used when transferring data from the digital video camera to the server.

Method

Description

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..... [3]



(d) Describe **two** reasons why the server that stores the videos uses magnetic hard disks instead of solid state (flash) memory.

1

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2

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[4]



- (e) The table contains three algorithms that perform data validation. The third algorithm uses the functions `MID` and `LENGTH`.

The functions `MID` and `LENGTH` are defined as follows:

- `MID(ThisString : STRING, x : INTEGER, y : INTEGER) RETURNS STRING`
returns a string of length `y` starting at position `x` from `ThisString`
- `LENGTH(ThisString : STRING) RETURNS INTEGER`
returns the integer value representing the length of `ThisString`

Complete the table to identify the method of data validation for each algorithm. Each method of data validation must be different.

Algorithm	Method of data validation
<pre>INPUT x IF x < 1 or x > 26 THEN OUTPUT "Invalid" ENDIF</pre>	
<pre>INPUT x IF x <> 'R' AND x <> 'G' AND x <> 'B' THEN OUTPUT "Invalid" ENDIF</pre>	
<pre>INPUT x FLAG ← FALSE FOR INDEX ← 1 TO LENGTH(x) IF MID(x, INDEX, 1) = "@" THEN FLAG ← TRUE ENDIF NEXT INDEX IF NOT FLAG THEN OUTPUT "Invalid" ENDIF</pre>	

Question 8 - (9618_w25_13_4)

Subsection Topics: (a) : 2.1 - Networks Including the Internet | (b) : 2.1 - Networks Including the Internet | (c) : 3.1 - Computers and Their Components

A network engineer is setting up a Local Area Network (LAN) in an office. The network uses both wired and wireless connectivity.

(a) Give **three** advantages of a wireless network compared to a wired network.

1

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2

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3

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[3]

(b) Give **one** drawback of using a wireless network.

.....

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[1]

(c) Complete the table by writing the name of the type of memory technology for each description.

Description	Memory technology
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	
A read only memory where the contents can be written only once after manufacture.	

[3]

Question 9 - (9618_s24_11_2)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 3.1 - Computers and Their Components, 4.1 - CPU Architecture | (d) : 1.2 - Multimedia | (e) : 2.1 - Networks Including the Internet

A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



(a) The video doorbell can be considered an example of an embedded system.

Identify **two** characteristics of the doorbell that suggest it is an embedded system.

- 1
-
- 2
-

[2]

(b) State whether the video doorbell is a monitoring system or a control system.
Justify your choice.

- Monitoring or control system
- Justification
-
-
-
-

[2]

(c) The video doorbell has both primary memory and secondary storage.

(i) Identify **two** items of data that the video doorbell will store in primary memory.

1

.....

2

.....

[2]

(ii) The video doorbell has a solid state (flash) secondary storage device.

Complete the table by writing the answer or answers to each statement about the principal operation of solid state (flash) memory.

Statement	Answer
the two types of logic gate that can be used to create solid state devices	1 2
the number of transistors contained in each cell	
the type of gate that can retain electrons without power	
the type of gate that allows or stops current from passing through	

[4]

(iii) The video doorbell uses a buffer.

Describe how the video doorbell will use the buffer.

.....

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.....

.....

[2]

(d) The digital camera has a microphone which is used to record the sound for the video.

The user changes the sampling rate that the microphone uses from 44.1 kHz to 88.2 kHz.

Describe how this change in sampling rate will affect the performance of the video doorbell.

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..... [3]

(e) The video doorbell allows both real-time and on-demand bit streaming.

(i) State what is meant by bit streaming.

.....

..... [1]

(ii) Give **two** differences between real-time and on-demand bit streaming.

1

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2

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..... [2]

Question 10 - (9618_s24_12_2)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 4.1 - CPU Architecture | (c) : 3.1 - Computers and Their Components | (d) : 1.2 - Multimedia, 1.3 - Compression

A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(a) Complete the description of the principal operation of a VR headset.

A headset can have one or two that output the image to the user. The headset has speakers that output surround sound to give a realistic experience.

The user's head movements are detected using a sensor.

This sensor is a The data is transmitted to a microprocessor that analyses the data to identify the of movement. Some headsets use that record the user's eye movements for analysis.

[4]

(b) The computer uses a buffer when transmitting data to the VR headset.

Explain how a buffer is used when data is transmitted between the computer and the VR headset.

[3]

[3]

(c) The VR headset has Electrically Erasable Programmable Read Only Memory (EEPROM).

Explain the benefits of using EEPROM instead of other types of Read Only Memory (ROM) in the VR headset.

..... [3]

[3]

(d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.

(i) Describe how the data for a bitmapped image is encoded.

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..... [3]

(ii) Describe the contents of a vector graphic drawing list.

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..... [2]

(iii) The bitmap video is **not** compressed before transmission to the VR headset.

Give **two** reasons why the video does **not** need to be compressed.

1

.....

2

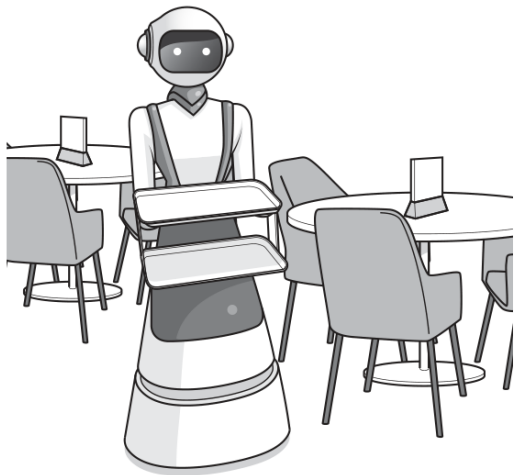
.....

[2]

Question 11 - (9618_s24_13_7)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 9.1 - Computational Thinking | (c) : 3.1 - Computers and Their Components | (d) : 3.1 - Computers and Their Components | (e) : 11.3 - Structured Programming | (f) : 6.2 - Data Integrity, 6.1 - Data Security

Robots are used to serve food and drink to customers at a restaurant.



(a) A robot navigates through the restaurant to the table it is serving.

Complete the table by identifying **two** sensors that can be included in the robot **and** the purpose of each sensor in the navigation system.

Sensor	Purpose of sensor in navigation system
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.....	

[2]

(b) The robot uses Artificial Intelligence (AI) to communicate with the customers. The customers speak to the robot to order their food and drinks.

Explain how AI will be used in this part of the robot.

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- (c) The navigation system can be considered an example of a control system.

Describe how feedback is used in a control system.

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..... [2]

- (d) The robot includes a touchscreen for the customer to make their payment.

Describe the principal operation of a touchscreen.

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..... [4]



(e) Program libraries were used when writing the robot's software.

(i) State what is meant by a **program library**.

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..... [1]

(ii) Some program libraries include Dynamic Link Library (DLL) files.

Describe the benefits of a programmer using a library with DLL files instead of using a library that does not include DLL files.

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..... [4]



- (f) The data from the robots is transmitted to a central computer using a wireless connection.
- (i) Complete the table by identifying **and** describing **two** methods of data verification that can be used during data transfer.

	Method	Description
1		
2		

[4]

- (ii) Explain how encryption can protect the security of data during transmission.

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[2]

Question 12 - (9618_w24_11_5)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.2 - Logic Gates and Circuits | (c) : 3.1 - Computers and Their Components

A security system has both a floodlight (very bright light) and an audio alarm.

The data from multiple sensors is analysed and used to:

- turn on the floodlight
- sound the audio alarm.

Sensors can be used to detect:

- if doors are open
- the external daylight level
- if people are detected within a set distance.

(a) Complete the table to identify the most appropriate type of sensor for each scenario.

Scenario	Type of sensor
A door is open.	
The external daylight level is below a set amount.	
A person is detected within 2 metres.	

[1]



(b) The floodlight (**X**) and audio alarm (**Y**) operate according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
A	external daylight level	1	Low
		0	High
B	front door	1	Open
		0	Closed
C	person is within 2 m	1	Detected
		0	Not detected
D	back door	1	Open
		0	Closed
E	security system	1	Switched on
		0	Switched off

The floodlight turns on (**X = 1**) if:

- the security system is switched on
and
- the external daylight level is low
and
- a person is detected within 2 m.

The audio alarm turns on (**Y = 1**) if:

- the security system is switched on
and
- one **or** more doors are open, **or** a person is detected within 2 m.

Write logic expressions for the security system.

X =

.....

Y =

.....

[2]

(c) Explain whether the security system is an example of a monitoring system or a control system.

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Question 13 - (9618_w24_11_3)

Subsection Topics: (a) : 4.1 - CPU Architecture | (b) : 4.1 - CPU Architecture, 3.1 - Computers and Their Components | (c) : 3.1 - Computers and Their Components | (d) : 3.1 - Computers and Their Components, 4.1 - CPU Architecture

A computer system has a dual-core Central Processing Unit (CPU).

(a) State the purpose of the system clock and the Control Unit (CU) in a CPU.

System clock
.....
CU
.....
[2]

(b) (i) The number of cores in the processor affects the performance of the computer system.

Identify **one other** feature of a processor that can affect the performance of a computer system **and** state why it affects the performance.

Feature
.....
Reason
.....
[2]

(ii) A solid state (flash) memory drive is automatically recognised by the computer when it is plugged into a port in the computer.

Identify an appropriate type of port to connect the solid state memory drive to the computer.

Explain how this port provides an automatic connection.

Port
Explanation
.....
.....
.....
[3]



(c) Identify **two** disadvantages of using Dynamic RAM (DRAM) instead of Static RAM (SRAM) in a computer system.

- 1
-
- 2
-
- [2]

(d) The computer system is used to store data received from a temperature sensor every five seconds. The data is stored on an optical disc using an optical disc reader/writer.

- (i) Describe the principal operation of an optical disc reader/writer.
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-
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-
-
-
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- [4]

- (ii) The computer uses a buffer when writing data to the optical disc.
- Explain the use of a buffer when writing data to the optical disc.
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-
-
-
-
-
- [3]

Question 14 - (9618_w24_12_2)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 3.1 - Computers and Their Components

Embedded systems are used in many electronic devices.

(a) Describe the drawbacks of embedded systems.

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..... [3]

(b) An embedded system has Dynamic RAM (DRAM).

Identify **one** benefit of using DRAM instead of Static RAM (SRAM) in an embedded system.

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..... [1]

(c) Give **two** differences between Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM).

Difference 1

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Difference 2

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..... [2]

Question 15 - (9618_w24_12_3)

Subsection Topics: (a) : 4.1 - CPU Architecture | (b) : 3.1 - Computers and Their Components | (c) : 5.1 - Operating Systems

A student has a computer.

(a) The computer is designed using the Von Neumann model for a computer system.

Complete the table by describing the purpose of each of the given registers.

Register	Purpose
Program Counter (PC)	
Memory Address Register (MAR)	
Memory Data Register (MDR)	
Index Register (IX)	

[4]



- (b) The student needs to connect the computer to a monitor that has a screen resolution of 2560×1600 pixels. The monitor also has built-in speakers.

The computer has a Video Graphics Array (VGA) port and a High Definition Multimedia Interface (HDMI) port.

Explain the benefits of connecting the monitor to the computer using the HDMI port instead of the VGA port.

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..... [4]

- (c) The computer has an Operating System (OS). One of the key management tasks of the OS is process management.

Describe the process management tasks performed by an OS.

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..... [4]

Question 16 - (9618_w24_12_9)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components

A road bridge has a weight limit and a height limit for vehicles. For example, a vehicle must weigh less than 10 000 kg and must have a height of less than 3 m.

The bridge has a warning system. If a vehicle is approaching the bridge and it exceeds one or both limits, a sign displays a warning telling the driver of the vehicle to stop.

(a) The bridge warning system uses sensors to detect if a vehicle exceeds the limits.

Complete the table by identifying **two** different sensors that could be used by the system **and** describe how each sensor is used by the system.

Sensor	Use in bridge warning system
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.....	
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.....	
.....	
.....	

[2]

(b) Explain whether the bridge warning system is an example of a monitoring system or of a control system.

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[2]

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Question 17 - (9618_w24_13_3)

Subsection Topics: (3) : 3.1 - Computers and Their Components

A car has an automatic braking system.

A sensor is used to measure the distance to objects that are in front of the car.

The car automatically brakes if an object is too close to the front of the car or the distance between the car and the object is decreasing too quickly. The closer the object is to the front of the car, the harder the car brakes so that the car slows down more quickly.

Explain the reasons why the automatic braking system of the car is a control system.

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..... [3]

Question 18 - (9618_w24_13_2)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 4.1 - CPU Architecture

A memory buffer uses Dynamic RAM (DRAM).

(a) Identify **two** differences between DRAM and Static RAM (SRAM).

Difference 1

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Difference 2

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[2]

(b) Explain how a memory buffer is used when a computer is transferring data to its magnetic hard disk drive.

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[4]

Question 19 - (9618_s23_11_4)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 9.1 - Computational Thinking | (d) : 2.1 - Networks Including the Internet

A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

(a) This computer has both Read Only Memory (ROM) and Random Access Memory (RAM).

(i) Describe the contents of the ROM in the central computer.

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..... [2]

(ii) The central computer has Dynamic RAM (DRAM).

Identify **two** advantages of using DRAM instead of Static RAM (SRAM).

1

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2

..... [2]

(b) The central computer stores the video files on secondary storage.

Describe **two** reasons why magnetic storage is more appropriate than solid state storage for this computer.

1

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2

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..... [4]

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- (c) The CCTV system uses Artificial Intelligence (AI) to identify the presence of a person in the house and to track their movements.

Describe how AI is used in this system.

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..... [3]

- (d) The CCTV cameras are connected to a network and transfer their data wirelessly to the central computer.

- (i) Each device on the network has an IP address.

Complete the description of IP addresses.

An IPv4 address contains groups of digits. Each group is represented in bits and the groups are separated by full stops.

An IPv6 address contains groups of digits. Each group is represented in bits. Multiple groups that only contain zeros can be replaced with a

[5]

- (ii) The network makes use of subnetting.

Describe **two** benefits of subnetting a network.

1

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2

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[4]

Question 20 - (9618_s23_12_5)

Subsection Topics: (a) : 4.1 - CPU Architecture | (b) : 4.1 - CPU Architecture | (c) : 3.1 - Computers and Their Components | (d) : 5.1 - Operating Systems

A student has purchased a new laptop.

(a) The laptop is designed using the Von Neumann model for a computer system.

Identify **two** types of signal that a control bus can transfer.

1

2

[2]

(b) Describe **two** ways the hardware of a laptop can be upgraded to improve the performance **and** explain how each upgrade improves the performance.

1

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2

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[4]

(c) Peripherals are connected to the laptop using ports.

(i) A printer is connected to a Universal Serial Bus (USB) port.

Describe how data is transmitted through a USB port.

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..... [1]

(ii) A monitor is connected to the laptop using a different type of port.

Identify **one** other type of port that can be used to connect the monitor.

..... [1]

(d) The laptop has systems software.

(i) Describe how the Operating System (OS) manages processes in the computer.

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..... [5]

(ii) Describe the purpose of utility software in a computer.

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..... [2]

Question 21 - (9618_s23_13_3)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 9.1 - Computational Thinking | (c) : 1.2 - Multimedia

A mobile telephone is used to record a video.

(a) The mobile telephone has a touchscreen. There are different types of touchscreen.

Complete the description of the principal operation of touchscreens.

A touchscreen has two layers. When the user touches the screen, the layers touch and a is completed.

A touchscreen has several layers. When the top layer is touched, there is a in the electric current.

A microprocessor identifies the of the touch.

[5]

(b) The mobile telephone uses a built-in digital camera to record the video.

The digital camera automatically focuses on the faces of people.

Explain how Artificial Intelligence (AI) is used by the camera to automatically focus on the faces of people.

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..... [3]

(c) The video includes a sound recording.

(i) Describe how sound is represented in a computer.

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.....

..... [3]

- (ii) A second video is recorded. The sound in the second video needs to be more precise.

Explain the reasons why increasing the sampling rate and the sampling resolution will improve the precision of the second recording.

Sampling rate

.....

.....

.....

Sampling resolution

.....

.....

.....

[4]

Question 22 - (9618_w23_11_9)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components

(a) Explain the importance of feedback in a control system.

.....

.....

.....

..... [2]

(b) Give **one** example of an embedded system **and** explain why it is an example of an embedded system.

Example

.....

Explanation

.....

.....

.....

.....

..... [3]

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- 1
-
- 2
-
- [2]

Question 24 - (9618_w23_12_1)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 3.1 - Computers and Their Components

A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

(a) An automated system counts the number of chocolate bars made at the end of production.

The system includes a sensor positioned above the conveyor belt.

Identify **one** appropriate type of sensor that can be used.

..... [1]

(b) A second automated system removes chocolate bars with an incorrect weight from the production line.

Describe the role of an **actuator** in this second system.

.....
.....
.....
..... [2]

(c) The factory has many different machines with embedded systems.

(i) Identify **two** features of embedded systems.

1
.....
2
..... [2]

(ii) Identify **one** drawback of embedded systems.

.....
..... [1]

Question 25 - (9618_w23_12_8)

Subsection Topics: (a) : 5.1 - Operating Systems | (b) : 5.1 - Operating Systems | (c) : 3.1 - Computers and Their Components, 4.1 - CPU Architecture

A computer has an Operating System (OS).

(a) State **one** purpose of the Operating System.

.....

..... [1]

(b) The Operating System has utility software.

The purpose of some utility software is to improve security.

Identify **one** example of utility software that is **not** intended to improve security.

Explain why this software is needed.

Utility software

Explanation

.....

.....

.....

..... [3]

(c) An optical disc reader/writer is connected to the computer.

(i) Give the name of **one** port that can provide a connection for the optical disc reader/writer.

..... [1]

(ii) Describe the roles of the address bus, the data bus **and** buffers in the process of writing data to the optical disc reader/writer.

.....

.....

.....

.....

.....

..... [3]

..... [4]

..

(d) Identify **one** port that could be used to connect the virtual reality headset to the laptop.

Justify your choice.

Port

Justification

.....

.....

.....

[3]

Question 27 - (9618_s22_11_2)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 3.2 - Logic Gates and Circuits

A computer has hardware and software.

(a) The hardware includes different types of memory.

(i) Complete the description of computer memory.

Random Access Memory (RAM) and Read Only Memory (ROM) are both examples of memory.

One item that is stored in RAM is

One item that is stored in ROM is

RAM can be either Static RAM (SRAM) or Dynamic RAM (DRAM).

SRAM uses transistors arranged as

DRAM uses transistors and

[5]

(ii) Explain the difference between Programmable ROM (PROM), Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM).

.....
.....
.....
.....
.....
..... [3]

—

..... [5]

[5]

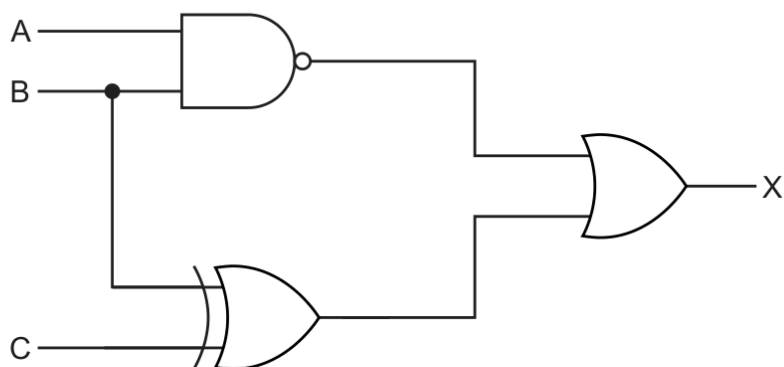
(c) Computers consist of logic gates.

- (i) Complete the table by writing **one** set of values (input 1 and input 2) for each gate that will give the output 1.

Gate	Input 1	Input 2	Output
AND			1
NAND			1
XOR			1
NOR			1

[4]

- (ii) Write the logic expression for the given logic circuit.



.....

.....

..... [3]

Question 28 - (9618_s22_12_2)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 3.1 - Computers and Their Components

A car has several features.

- (a) One feature is a lane detection system. This system monitors the lines on either side of the lane. If the car gets too close to one line, the system automatically moves the car away from the line.

Explain why the lane detection system is an example of an embedded system.

.....

.....

.....

..... [2]

- (b) Two other features:

- record the number of miles travelled in the current journey, from when the engine is turned on to when it is turned off
- record the total number of miles the car has travelled since it was built.

Identify the data that will be stored in the primary **and** secondary storage of the car for these **two** features.

Primary

.....

Secondary

..... [2]

- (c) The car has a resistive touchscreen for the user to select options.

Tick (✓) **one** box in each row to show whether each statement about a resistive touchscreen is true or false.

Statement	True	False
The screen always has five different layers		
A processor determines the horizontal and vertical coordinates of the point of contact		
The touchscreen will work if any object touches the screen		

[1]

[5]

Question 30 - (9618_w22_11_7)

Subsection Topics: (a) : 11.3 - Structured Programming | (b) : 5.1 - Operating Systems | (c) : 1.3 - Compression | (d) : 4.1 - CPU Architecture | (e) : 3.1 - Computers and Their Components

(a) State **two** benefits to a programmer of using Dynamic Link Library (DLL) files.

1

.....

2

.....

[2]

(b) Memory management is one of the tasks performed by an Operating System (OS).

Describe the ways in which memory management organises and allocates Random Access Memory (RAM).

.....

.....

.....

.....

[2]

(c) An Operating System may include a utility program to compress text files.

Describe **one** appropriate method of compressing a text file.

.....

.....

.....

.....

.....

.....

[3]

(d) Explain the reasons why increasing the amount of cache memory can improve the performance of a CPU.

.....

.....

.....

.....

[2]

(e) State the name of a peripheral device port that provides a physical connection in the computer for each of these peripherals.

3D printer

Monitor

Question 31 - (9618_w22_11_9)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components

Many modern televisions are examples of embedded systems.

(a) Explain why these televisions are embedded systems.

.....

.....

.....

..... [2]

(b) Embedded systems use Electrically Erasable Programmable ROM (EEPROM).

Describe **one** benefit of using EEPROMs in an embedded system.

.....

.....

.....

..... [2]

Question 32 - (9618_w22_12_3)

Subsection Topics: (a) : 3.2 - Logic Gates and Circuits | (b) : 3.2 - Logic Gates and Circuits | (c) : 3.1 - Computers and Their Components

(a) A greenhouse has an automatic window.

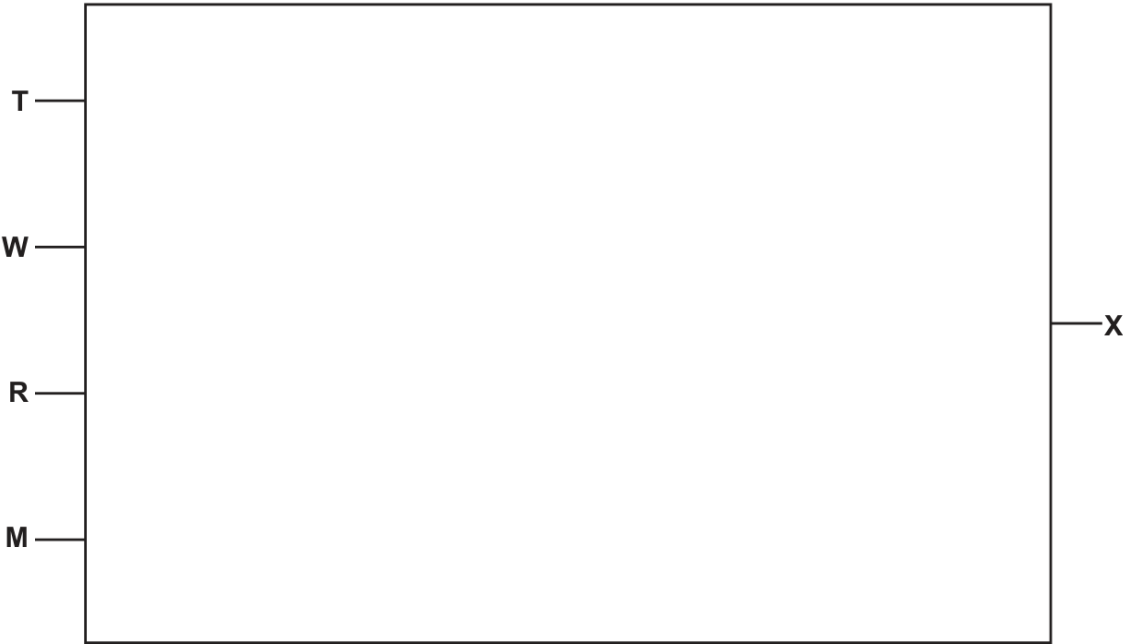
The window (X) operates according to the following criteria:

Parameter	Description of parameter	Binary value	Condition
T	Temperature	1	Too high
		0	Acceptable
W	Wind speed	1	Too high
		0	Acceptable
R	Rain	1	Detected
		0	Not detected
M	Manual override	1	On
		0	Off

The window opens (X = 1) if:

- the temperature is too high **and** the wind speed is acceptable
- and**
- rain is not detected, **or** the manual override is off.

Draw a logic circuit to represent the operation of the window.



(b) Complete the truth table for the logic expression:

$$X = \text{NOT } (A \text{ OR } B \text{ OR } C) \text{ AND } (B \text{ NOR } C)$$

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

[2]

(c) Embedded systems contain Read Only Memory (ROM) and Random Access Memory (RAM).

Explain the reasons why ROM is used in an embedded system.

.....

.....

.....

.....

.....

..... [2]

Question 33 - (9618_w22_13_10)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components

(a) Explain the importance of feedback in a control system.

.....

.....

.....

.....

.....

..... [3]

(b) (i) Identify **one** sensor that could be used in a car alarm system.

Justify your choice.

Sensor

Justification

.....

.....

..... [2]

(ii) The car alarm is an example of an embedded system.

Describe the characteristics of an embedded system.

.....

.....

.....

.....

.....

..... [3]

Question 34 - (9618_s21_11_5)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 3.1 - Computers and Their Components

Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

.....

.....

.....

..... [2]

(b) The washing machine’s embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine’s embedded system.

RAM

.....

ROM

..... [2]

(c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

.....

.....

.....

.....

.....

..... [2]

Question 35 - (9618_s21_13_5)

Subsection Topics: (a) : 3.1 - Computers and Their Components | (b) : 3.1 - Computers and Their Components | (c) : 3.1 - Computers and Their Components

Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

.....

.....

.....

..... [2]

(b) The washing machine’s embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine’s embedded system.

RAM

.....

ROM

..... [2]

(c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

.....

.....

.....

.....

.....

..... [2]

Answer for Question 1 (9618_s25_11_4)

Question	Answer	Marks						
4(a)	1 mark each to max 4 e.g. • Uses image recognition / facial recognition • Measures the distance between facial features of the customer • and stores the values in a database of user information • An image is captured using the digital camera • the AI identifies that the image is a face • by analysing the pixels to find patterns • the distance between features is calculated and compared to those in database	4						
4(b)	1 mark for sensor, 1 mark for corresponding use e.g. <table><tr><th>Sensor</th><th>Use</th></tr><tr><td>Pressure</td><td>Detects when the pressure of an item is removed from a shelf / put back on a shelf</td></tr><tr><td>Infrared</td><td>Detects when the beam is broken for an item removed / added</td></tr></table>	Sensor	Use	Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf	Infrared	Detects when the beam is broken for an item removed / added	2
Sensor	Use							
Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf							
Infrared	Detects when the beam is broken for an item removed / added							

Answer for Question 2 (9618_s25_11_6)

Question	Answer	Marks
6(a)(i)	1 mark for each correctly completed space • platters • spindle • read/write head • magnetic field The magnetic hard disk has one or more platters that can be magnetised. These are mounted on a spindle and rotate at high speed. A read/write head is moved across the surface on an arm. When data is read the changes in the magnetic field produce a change in the electric current.	4
6(a)(ii)	1 mark each to max 3 • More RAM means more currently running data and instructions can be stored • without needing to use virtual memory • without having to fetch the data from secondary storage first • which has a slower access time • Less latency / delay waiting for instructions / data	3
6(a)(iii)	1 mark each to max 2 e.g. • Wider data bus means more data can be transferred between components at a time • there is less delay / latency when fetching data for a running process • Wider address bus means larger memory addresses can be used • allowing more memory locations to be accessed directly • so less likely to run out of memory	2
6(b)	1 mark each to max 5 e.g. • To provide a user interface • so that the user is able to communicate with the hardware • To manage memory • so that data can be stored and accessed • and multitasking is possible • To manage files • allowing the user to create, edit, update and delete files and folders • To manage inputs and outputs from hardware/peripherals • To handle processes • to make sure each process has fair access	5

Answer for Question 3 (9618_w25_11_8)

Question	Answer	Marks
8(a)	1 mark for each correct explanation, max 2 marks Virus: Malicious program/software that replicates/copies itself and deletes/alters files/data stored on a computer Pharming: Malicious code/software installed on a computer which redirects user to a fake website to obtain personal data	2
8(b)	1 mark per bullet point, max 3 marks e. g. • Students should consider privacy issues • Students should not make inappropriate use of social networking // Students should not engage in cyberbullying • Students should not copy work that is not their own • Students should abide by the school rules/guidelines for accessing the network • Students should only use the school network for legitimate purposes • Students should not hack / crack other computers	3
8(c)	1 mark per bullet point, max 4 marks e. g. Negative Impacts • If students use AI to shortcut learning, they may miss out on developing reasoning and problem-solving skills // Students may not be learning anything // It could lead to a decline in specific skills • It could have negative impacts on student communication // heavy reliance on AI may limit opportunities for collaboration, teamwork, and face-to-face communication • Emphasises the digital divide// some students may not have access to the technology • The AI answer might not be correct leading to mis-information • Increased screen time and isolation could contribute to anxiety or loneliness Positive Impacts • Students given additional support may do better • Pupils who struggle in traditional settings may gain self-esteem through AI-assisted learning at their own pace Long-Term Impacts • Protection of intellectual property // ethical issues and considerations • Schools may need to redesign curricula and/or assessment to integrate AI	4

Answer for Question 5 (9618_w25_12_8)

Question	Answer	Marks
8(a)(i)	1 mark per bullet point, max 2 marks • Cloud storage can be free of charge for a limited amount • There is no need for personal storage devices • The data can be accessed from any computer with an internet connection • Cloud data services usually include data backup and recovery services • It is easier to share data / collaborate with colleagues • Cloud storage capacity can be quickly and easily increased if needed	2
8(a)(ii)	1 mark per bullet point, max 2 marks • The storage/cloud can only be accessed with an internet connection • There may be no backups // possibly no control over backups // reliance on third party for backups • Uploading / downloading data could take a long time • Potential issues with the cloud storage company impacting access e.g. downtime or technical problems • Potential compatibility issues if wanting to move the data • Limited storage availability without paying extra • ... meaning (possibly) more expensive in the long term • Possibly security issues	2
8(b)(i)	1 mark per bullet point, max 4 marks Methods to determine point of touch (Max 2) • Resistive // two layers of material make contact and complete a circuit • Capacitive // contact with the screen creates a change in charge • Infrared // the beams are broken by the pointing device • Optical imaging // a shadow is created when the surface is touched • Acoustic pulse // the acoustic wave is absorbed by the pointing device • Selection from the menu (Max 2) • The point of touch is used to determine the x and y coordinates • The menu item corresponding to the coordinate position is recognised and the menu item is added to the order	4
8(b)(ii)	1 mark per bullet point, max 2 marks • Disk repair / disk contents analysis • Back-up software	2

Answer for Question 6 (9618_w25_12_11)

Question	Answer	Marks
11	1 mark per bullet point, max 3 marks Identification: Control system (no mark) Justification: • The system uses an actuator to open or close the doors • The output / opening the doors allows people to move, which changes the input to the sensor • The system acts autonomously on the feedback from a sensor	3

Answer for Question 7 (9618_w25_13_7)

Question	Answer	Marks
7(a)	1 mark per point, max 2 marks e.g. • The learning experience could be improved • ... by identifying students who are struggling • and employing more personalised learning • There could be early Intervention • ... to determine which students require extra support • Teachers and students could understand the challenges better • ...and identify optimal times for learning • There could be privacy concerns • students / teachers could be uncomfortable with constant surveillance • ... students might not want their data / actions passed to third parties • There could be Mental Health concerns • due to constant monitoring and pressure • contributing to student stress and anxiety	2
7(b)	1 mark for the working 1 mark for the correct answer Working: $(4000 \times 3000 \times 30 \times 16) / (8 \times 1000 \times 1000 \times 1000)$ $//(4000 \times 3000 \times 30 \times 2) / (1000 \times 1000 \times 1000)$ Answer: 0.72 gigabytes	2
7(c)	1 mark for the name 2 marks for the matching description, max 3 marks • Parity Byte Check • a parity bit is added to each byte to make the number of 1s match the parity, odd or even • each byte can be checked on receipt and request to be resent if the byte does not match parity • Parity Block Check • a bit is added to each byte, but a parity byte is also set for each block • the location of an error can be found using vertical and horizontal parity • Checksum • A calculation is made from the data and transmitted with the data • The receiver performs the same calculation and compares with received checksum to see if they match	3

Answer for Question 7 (9618_w25_13_7)

Question	Answer	Marks
7(d)	1 mark per reason, 1 mark for corresponding justification x2 max 4 marks • Lower cost per unit of storage • ... so the high capacity of storage required for large number of video files will be less costly • A large number of read/write operations are being performed continuously • ... and magnetic storage is likely to have a longer life span compared with solid state	4
7(e)	1 mark for each validation check correctly identified, max 3 marks Range Check Existence Check Format Check	3

Answer for Question 8 (9618_w25_13_4)

Question	Answer	Marks
4(a)	1 mark for each benefit, max 3 • Devices can be more mobile as no physical connection is required • No cabling is needed so easier to set up • Easier to add additional devices • Multiple different device types can be connected	3

Answer for Question 8 (9618_w25_13_4)

Question	Answer	Marks								
4(b)	1 mark for a drawback • Can be less secure • Transmission speed can be slow • Interference is possible • The signal degrades quickly without repeaters/boosters	1								
4(c)	One mark for each correct row <table><tr><th>Description</th><th>Memory Technology</th></tr><tr><td>A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.</td><td>EEPROM</td></tr><tr><td>A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.</td><td>EPROM</td></tr><tr><td>A read only memory where the contents can be written only once after manufacture.</td><td>PROM</td></tr></table>	Description	Memory Technology	A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	EEPROM	A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	EPROM	A read only memory where the contents can be written only once after manufacture.	PROM	3
Description	Memory Technology									
A read only memory where the contents can be erased using a voltage pulse. The contents can be changed multiple times without physically removing the memory.	EEPROM									
A read only memory where the contents are erased using ultraviolet (UV) light. The memory must be physically removed to be reprogrammed.	EPROM									
A read only memory where the contents can be written only once after manufacture.	PROM									

Answer for Question 9 (9618_s24_11_2)

Question	Answer	Marks										
2(c)(ii)	1 mark for each row: <table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The two types of logic gate that can be used to create solid state devices</td><td>NAND NOR</td></tr><tr><td>The number of transistors contained in each cell</td><td>2</td></tr><tr><td>The type of gate that can retain electrons without power</td><td>floating</td></tr><tr><td>The type of gate that allows or stops current from passing through</td><td>control</td></tr></table>	Statement	Answer	The two types of logic gate that can be used to create solid state devices	NAND NOR	The number of transistors contained in each cell	2	The type of gate that can retain electrons without power	floating	The type of gate that allows or stops current from passing through	control	4
Statement	Answer											
The two types of logic gate that can be used to create solid state devices	NAND NOR											
The number of transistors contained in each cell	2											
The type of gate that can retain electrons without power	floating											
The type of gate that allows or stops current from passing through	control											
2(c)(iii)	1 mark each to max 2: • Captured video is transmitted to buffer • ... video is transmitted from buffer to smartphone • Store recent data in a buffer for the user to rewind • ... instead of storing everything in secondary storage • Store readings from motion sensor • ... until the microprocessor can process them • Store video from digital camera • ... before moving it to secondary storage	2										
2(d)	1 mark each to max 3: • Data transmission to user's smartphone will take longer • ... because there is more data to transmit • The secondary storage device will fill faster • ... fewer videos will be able to be stored long-term // videos are overwritten more often	3										
2(e)(i)	1 mark for: Continuous ordered flow of bits over a communication path	1										
2(e)(ii)	1 mark each to max 2: • Real-time is direct from source whereas on-demand is pre-recorded/downloaded to view later • Real-time cannot be re-watched, on-demand can be paused, re-watched etc. • Real-time plays continually, on-demand downloads sections/blocks and cannot play until next section is downloaded	2										

Answer for Question 10 (9618_s24_12_2)

Question	Answer	Marks
2(a)	1 mark for each correctly completed statement: • (LCD) displays/screens/lenses • gyroscope/accelerometer • direction/speed • digital cameras A headset can have one or two (LCD) displays/screens/lenses that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor. This sensor is a gyroscope/accelerometer. The data is transmitted to a microprocessor that analyses the data to identify the direction/speed of movement. Some headsets use digital cameras that record the user's eye movements for analysis.	4
2(b)	1 mark each to max 3: • The buffer is used as a temporary store for data going to the headset • Data is transferred into the buffer by the computer • Data is retrieved from the buffer by the headset • When the buffer is empty/full an interrupt is sent to the computer requesting more data/stopping further data being sent • When the headset has enough data/needs more data, an interrupt is sent by the headset to the computer to stop sending data from buffer	3
2(c)	1 mark each to max 3: • EEPROM allows frequent/multiple read/write/erase operations • ... so the headset can take advantage of new features • ... without fully erasing the contents of the firmware in the headset first // can erase a particular byte or the whole EEPROM • ... without removing the chip(s)/firmware from the headset • ... the contents of the firmware in the headset can be changed by the user without technical expertise • Cheaper to manufacture so headset will be cheaper to purchase	3
2(d)(i)	1 mark each: • Image is made of pixels and each pixel has one colour • Each colour has a unique binary code • Code for the colour of each pixel is stored in sequence	3
2(d)(ii)	1 mark each to max 2: • List of objects in the drawing • A list that stores the command/description/equation required to draw each object • Properties of each object e.g. the fill colour, line weight/colour	2

Answer for Question 10 (9618_s24_12_2)

Question	Answer	Marks
2(d)(iii)	1 mark each to max 2: • Dedicated connection to the headset // not sharing bandwidth • Already fast connection that can transmit the data without slowing • Video may already be a small file size and does not need further reduction • Video is not saved so storage is not an issue in the headset	2

Answer for Question 11 (9618_s24_13_7)

Question	Answer	Marks								
7(a)	1 mark for sensor and matching purpose to max 2: <table><tr><th>Sensor</th><th>Purpose of sensor in navigation system</th></tr><tr><td>Pressure</td><td>To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on</td></tr><tr><td>Infra-red</td><td>To detect if there is an obstacle in the way // to indicate that it has reached the desired table</td></tr><tr><td>Sound</td><td>To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop</td></tr></table>	Sensor	Purpose of sensor in navigation system	Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on	Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table	Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop	2
Sensor	Purpose of sensor in navigation system									
Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on									
Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table									
Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop									
7(b)	1 mark each to max 3: e.g. • Voice/speech recognition is used • ... to identify if someone speaking • The sound is recorded and analysed • The audio recordings are compared to a database of words/sound waves • ... to identify the word that has the highest probability of being said • Natural language recognition is used • Words are combined and compared to known sentences • ... programmed action(s) for matching sentence(s) are performed	3								
7(c)	1 mark each to max 2: • Feedback ensures that a system operates within set criteria / constraints • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be automatically adjusted	2								
7(d)	1 mark each to max 4: • Resistive: The space between the conductive layers is removed/the layers touch and a circuit is completed • Capacitive: The electrical charge changes where the user pressed • The point of contact is identified • ... from the change in electrical field • The software/microprocessor calculates the coordinates	4								
7(e)(i)	1 mark for: Pre-written code/functions/routines that can be imported/called in another program	1								

Answer for Question 11 (9618_s24_13_7)

Question	Answer	Marks								
7(e)(ii)	1 mark each to max 4: • Maintenance not needed to be done by the programmer • ... because the DLL is separate from program • The calling program does not need recompilation by the programmer when a DLL file changes • ... because the DLL file can be updated independently of the calling program • ... updates will apply to all programs that use the DLL file	4								
7(f)(i)	1 mark for each correct method and 1 mark for corresponding description to max 4: <table><tr><th>Method</th><th>Description</th></tr><tr><td>Parity byte</td><td>An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.</td></tr><tr><td>Parity block</td><td>Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.</td></tr><tr><td>Checksum</td><td>A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.</td></tr></table>	Method	Description	Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.	Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.	Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.	4
Method	Description									
Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.									
Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.									
Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.									
7(f)(ii)	1 mark each to max 2: • Encodes/scrambles data • ... so if it is intercepted it cannot be understood • Algorithm/key is required to decode the data	2								

Answer for Question 12 (9618_w24_11_5)

Question	Answer	Marks								
5(a)	1 mark for all 3 appropriate sensors <table><tr><th>Scenario</th><th>Type of sensor</th></tr><tr><td>A door is open.</td><td>Pressure sensor // Infra-red</td></tr><tr><td>The external daylight-level is below a set amount.</td><td>Light sensor</td></tr><tr><td>A person is detected within 2 metres.</td><td>Infra-red</td></tr></table>	Scenario	Type of sensor	A door is open.	Pressure sensor // Infra-red	The external daylight-level is below a set amount.	Light sensor	A person is detected within 2 metres.	Infra-red	1
Scenario	Type of sensor									
A door is open.	Pressure sensor // Infra-red									
The external daylight-level is below a set amount.	Light sensor									
A person is detected within 2 metres.	Infra-red									
5(b)	1 mark for each correct expression X = E AND A AND C Y = E AND (B OR C OR D)	2								
5(c)	1 mark for each bullet point (max 3) for a correct justification No marks for the identification of the system Monitoring system • There is no use of feedback // The light and the alarm are just warnings • The output of the floodlight or the audio alarm does not affect the input of data from the sensors • The system does not have any actuators	3								

Answer for Question 13 (9618_w24_11_3)

Question	Answer	Marks
3(a)	1 mark for system clock and 1 mark for Control Unit System clock • To synchronise operations • ... by creating and transmitting timing signals on the control bus Control Unit • Sends/receives control signals along control bus • Reads an instruction from the contents of the memory location whose address is stored in PC • Coordinates/synchronises the activity of other components in the CPU • Manages the execution of instructions • Controls communication between the components in the CPU	2
3(b)(i)	1 mark for the feature, 1 mark max for the matching reason e.g. Feature: clock speed • Higher clock speed means that more F-E cycles are executed per second // Higher clock speed results in more throughput Feature: bus width • Larger bus width means that more data transferred at the same time	2

Answer for Question 13 (9618_w24_11_3)

Question	Answer	Marks
3(b)(ii)	1 mark for correct port. 2 marks for explanation Port: USB / Universal Serial Bus Explanation: • A voltage change occurs when the drive is plugged in • The computer detects this voltage change • The code of the device is transferred to computer • ... the OS finds the code of the device in the list of devices • ... and loads the appropriate device driver	3
3(c)	1 mark for each bullet point (max 2) e.g. • DRAM requires constant refresh cycles unlike SRAM • DRAM has lower access speed than SRAM	2
3(d)(i)	1 mark for each bullet point (max 4) • The disc is spun at high speed • A laser is shone onto the disc to read / write • ... using optical head to move it into position • ... it follows the spiral track from the centre outwards • When writing the laser burns pits to represent the data • When reading the laser reflects from pits and lands • The reflection from a pit and a land is different • ... the differences are interpreted as 1 or 0	4
3(d)(ii)	1 mark for each bullet point (max 3) • The computer and optical disc reader/writer send and receive at different speeds • A buffer allows temporary storage of the data • ... so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer // so the computer can carry on with other tasks • ... so the optical disc reader/writer is not overloaded • ... and so that data is transferred to optical disc reader/writer from the buffer at the slower rate	3

Answer for Question 14 (9618_w24_12_2)

Question	Answer	Marks
2(a)	1 mark for each bullet point (max 3) e.g. • It is difficult to change / update the firmware by the user • Errors cannot be fixed easily // Troubleshooting / fault-finding / repairing is a specialist task / expensive • Functionality cannot be changed/extended easily // Cannot be easily adapted for another task • Faulty / outdated devices are often thrown away rather than repaired • ... leading to e-waste	3

Answer for Question 14 (9618_w24_12_2)

Question	Answer	Marks
2(b)	1 mark from: • DRAM is less expensive to manufacture/purchase than SRAM in the embedded system • DRAM has a higher bit density per chip // more data can be stored per chip	1
2(c)	1 mark for each bullet point (max 2) • EPROM uses ultraviolet light to erase data whilst EEPROM uses an electrical signal to do this • EPROM has to be removed from the circuit board when changing the data whilst EEPROM remains in the circuit when the data is changed • EPROM erases all the data, EEPROM can erase parts of the data	2

Answer for Question 15 (9618_w24_12_3)

Question	Answer	Marks										
3(a)	1 mark for each correct purpose <table><tr><th>Register</th><th>Purpose</th></tr><tr><td>Program Counter (PC)</td><td>Stores the address of the next instruction to be fetched/executed</td></tr><tr><td>Memory Address Register (MAR)</td><td>Stores the address of the memory location where data will be read from/written to</td></tr><tr><td>Memory Data Register (MDR)</td><td>Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR</td></tr><tr><td>Index Register (IX)</td><td>Stores a number that will be added to the operand, to form the address of the data</td></tr></table>	Register	Purpose	Program Counter (PC)	Stores the address of the next instruction to be fetched/executed	Memory Address Register (MAR)	Stores the address of the memory location where data will be read from/written to	Memory Data Register (MDR)	Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR	Index Register (IX)	Stores a number that will be added to the operand, to form the address of the data	4
Register	Purpose											
Program Counter (PC)	Stores the address of the next instruction to be fetched/executed											
Memory Address Register (MAR)	Stores the address of the memory location where data will be read from/written to											
Memory Data Register (MDR)	Stores the data read from the address in the MAR // stores the data to be written to the address in the MAR											
Index Register (IX)	Stores a number that will be added to the operand, to form the address of the data											
3(b)	1 mark for each bullet point (max 4) e.g. • HDMI has faster transfer rates than VGA • ... needed due to high resolution / large number of pixels of monitor // HDMI supports the high resolution of the monitor • HDMI supports video and audio transfer between computer and monitor speakers • ... so no separate sound cable is needed unlike VGA • HDMI is digital interface therefore no data is lost in transfer to analogue and back • HDMI is less prone to error/crosstalk/external interference	4										

Answer for Question 15 (9618_w24_12_3)

Question	Answer	Marks
3(c)	1 mark for each bullet point (max 4) e.g. • It manages the scheduling of processes // Decides which process is to be run next • Allows multi-tasking/multi-processing • Ensures fair access • Handles interrupts • Manages / allocates which resources the processes require • Facilitates the sharing and exchange of data between processes • Prevents interference between processes // conflict resolution	4

Answer for Question 16 (9618_w24_12_9)

Question	Answer	Marks
9(a)	1 mark for sensor and its use in this system • Infra-red sensor • Measure / check the height of the vehicle • Pressure sensor • Measure / check the weight of the vehicle	2
9(b)	1 mark for each bullet point (max 2) for a correct justification No marks for the identification of the system Monitoring system • Because there is no use of feedback // the warning sign is only an indicator • ... the output of the turning on of the sign does not affect the input of data from the sensors • The system does not have any actuators	2

Answer for Question 17 (9618_w24_13_3)

Question	Answer	Marks
3	1 mark for each bullet point (max 3) • The system uses feedback • The system acts independently • Input data causes the car to brake • ... braking decreases the/changes the distance to the object • ... this new distance is used to determine whether/how hard the car needs to brake	3

Answer for Question 18 (9618_w24_13_2)

Question	Answer	Marks
2(a)	1 mark for each difference e.g. • DRAM requires constant refreshing, but SRAM does not need to be constantly refreshed • DRAM generally has greater capacities per chip, but SRAM generally has lower capacities per chip • DRAM has slower access times than SRAM	2
2(b)	1 mark for each bullet point (max 4) • The computer and the hard disk drive transmit and receive at different speeds // The computer transfers data faster than the HDD can receive • The buffer is used for temporary storage • ... so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer • ... and so that data is transferred to hard disk drive from the buffer at the slower rate	4

Answer for Question 19 (9618_s23_11_4)

Question	Answer	Marks
4(a)(i)	1 mark each to max 2 • Stores the bootstrap program // start-up instructions for the central computer // BIOS • Stores the start-up instructions for the CCTV system/cameras // firmware for CCTV • Stores the kernel of the Operating System // stores parts of the Operating System	2
4(a)(ii)	1 mark each to max 2 • Costs less per unit • Higher storage density • Simple design – uses fewer transistors	2
4(b)	1 mark for reason, 1mark for application/justification • The computer will have a large number of read/write operations because it is working all the time • .. magnetic storage has more longevity • Magnetic storage costs less per storage unit • ... videos are large files and therefore very large storage capacity is required	4
4(c)	1 mark each to max 3 Examples: • Uses image recognition • Monitors every image taken to identify matching images/shapes/features to a 'person' ... • ... starts recording to secondary storage/permanently when a person is identified • System identifies direction of movement of person and uses this to decide where/how to move the camera/record • System identifies other cameras to start recording based on direction of movement	3
4(d)(i)	1 mark for each term An IPv4 address contains 4 groups of digits. Each group is represented in 8 bits and the groups are separated by full stops. An IPv6 address contains 8 groups of digits. Each group is represented in 16 bits. Multiple groups that only contain zeros can be replaced with a :: // double colon.	5

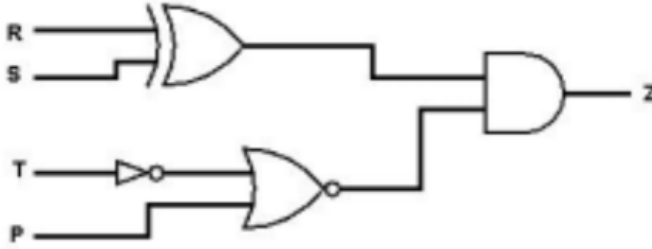
Answer for Question 19 (9618_s23_11_4)

Question	Answer	Marks
4(d)(ii)	1 mark for identification, 1mark for expansion e.g. • Reduce amount of traffic in a network // improve network speed • Data stays in its subnet so it does not travel as far • Improves network security • .. so that not all devices can access all areas of the network • Allows for easier maintenance • ... because only one subnetwork may need taking down/changing while the rest of the network can continue	4

Answer for Question 20 (9618_s23_12_5)

Question	Answer	Marks
5(a)	1 mark each to max 2 Examples: • Interrupt • Timing • Read • Write	2
5(b)	1 mark for description; 1 mark for corresponding explanation Examples • Increase number of cores • Each core can independently carry out a process at the same time // so that more instructions are performed in parallel • Increase RAM capacity • ... allowing more applications to reside in memory at the same time, saving disk access times • Increase cache memory • More data can be stored in fast access so less time is spent accessing from RAM • Increase clock speed • More Fetch-Decode-Execute (FDE) cycles can run each second / per unit time	4
5(c)(i)	1 mark for a correct answer • 1 bit is transferred at a time • Can be synchronous or asynchronous • USB-3 is full duplex and earlier versions are half-duplex	1
5(c)(ii)	1 mark for identification of a suitable port Examples • HDMI • DisplayPort	1

Answer for Question 20 (9618_s23_12_5)

Question	Answer	Marks																																													
5(d)(i)	1 mark each to max 5 • Manages the scheduling of processes // decides which order to run processes • Manages which resources the processes require • ... such as allocating memory • Enables processes to share data • Prevents interference between processes // resolution of conflicts • Handles the process queue • It allows multi-tasking / multi-processing • ... by ensuring fair access, handling priorities and handling interrupts	5																																													
5(d)(ii)	1 mark each to max 2 • To help users to set-up / configure / analyse / optimise / maintain the computer ... • ... by for example, making memory allocation more efficient • ... by for example, checking the system for faults	2																																													
6(a)	1 mark for correct XOR and AND gates, with correct inputs 1 mark for correct NOT and NOR gates with correct inputs 	2																																													
6(b)	1 mark for first 4 rows 1 mark for last 4 rows <table><tr><th>P</th><th>Q</th><th>R</th><th>Working space</th><th>Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	P	Q	R	Working space	Z	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		0	1	1	0		1	1	1	1		1	2
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Answer for Question 21 (9618_s23_13_3)

Question	Answer	Marks
3(a)	1 mark for each term • Resistive • Circuit • Capacitive • Change • Coordinates A resistive touchscreen has two layers. When the user touches the screen, the layers touch and a circuit is completed. A capacitive touchscreen has several layers. When the top layer is touched there is a change/increase/decrease in the electric current. A microprocessor identifies the coordinates of the touch.	5
3(b)	1 mark each to max 3 Examples: • Scans the scene in real time • Identifies if there are faces in the image • Uses facial recognition • ... uses image recognition • ... takes each frame individually • ... analyses the pixels • ... stores pattern for a face • ... looks for patterns that match/come close to the pattern for a face • Camera focuses on the pattern identified	3
3(c)(i)	1 mark each • The amplitude is recorded a set number of times a second • Each (instance of an) amplitude is given a corresponding binary number • The binary number (of each amplitude) is saved in sequence	3
3(c)(ii)	1 mark each; max 2 for rate and max 2 for resolution Sampling rate • There are smaller 'gaps' in the sound wave // sound is recorded more often • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller Sampling resolution • There are more bits per sample // a wider range of amplitudes can be stored • Each binary amplitude/note (in the digital recording) is closer to the analogue amplitude/note • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller	4

Answer for Question 22 (9618_w23_11_9)

Question	Answer	Marks
9(a)	One mark for each bullet point (max 2) • Feedback ensures that a system operates within set criteria / constraints • ...by enabling system output to affect (subsequent) system input • ...thus allowing conditions to be <u>automatically</u> adjusted	2
9(b)	One mark for each reason to max 3 to match the example given • Dedicated to one task applied to example • Does not require much processing power applied to example • Built into a larger system applied to example • Contains firmware that cannot be easily updated applied to example • The system does not have its own operating system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	3

Answer for Question 23 (9618_w23_11_7)

Question	Answer	Marks
7(a)	1 mark for each bullet point (max 3) Generic mark points: • Additive manufacturing • Uses a digital 3D model or a Computer Aided Design (CAD) (file) • Builds up the model one layer at a time • ...starting from the bottom • ...using x, y and z co-ordinates • The material is fused / cured together layer by layer Specific mark points: Fused Deposition Modelling (FDM) • Material is heated and pushed through nozzle / extruder Stereolithography (SLA) • Photosensitive liquid resin is exposed to a UV-laser beam Digital Light Processing (DLP) • Uses liquid plastic resin melted with arc lamps Selective Laser Sintering (SLS) • Uses a laser to form objects from powdered material	3
7(b)	1 mark for each bullet point (max 2) • To prevent overheating // ensure material is hot enough • ...by identifying the temperature of the object (being printed) • ...by identifying the temperature of the material being used	2
7(c)	1 mark for each bullet point (max 2) • Dynamic RAM has lower cost per unit • A fast access speed is not needed • Higher bit density // more data can be stored per chip	2

Answer for Question 24 (9618_w23_12_1)

Question	Answer	Marks
1(a)	1 mark for: infra-red / proximity (sensor)	1
1(b)	1 mark for each bullet point (max 2) - Actuator generates a signal / causes an action / converts electrical energy into a mechanical force ... to push an arm // to open a trap door // to pick up the chocolate bar with the incorrect weight	2
1(c)(i)	1 mark for each feature (max 2) - Dedicated to a single task // limited number of functions - Built into a larger system // integrated into a larger system - An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	2
1(c)(ii)	1 mark for each bullet point (max 1) - Difficult to change / update the firmware by the user // Difficult to upgrade devices to take advantage of new technology - Cannot be easily adapted for another task - Troubleshooting faults/upgrading is a specialist task - Devices are often thrown away rather than repaired as difficult to upgrade or repair	1

Answer for Question 25 (9618_w23_12_8)

Question	Answer	Marks
8(a)	1 mark for each bullet point (max 1) • To hide the complexities of the hardware from the user • To provide a platform for software to run • To provide a user interface	1
8(b)	1 mark for the name of the utility software 2 marks for the explanation • Defragmentation software • ... because over time saving and deleting of small files fragments the disk • ... the software makes (individual) files contiguous • ... so access time to the files is improved • ... because head movement is reduced • Disk contents analysis/disk repair software • ... to identify and mark bad sectors • ... to restore corrupted files • ... to recover lost data (due to hardware failure) • File compression • ... to reduce the size of files • ... which saves storage and memory space // by example • ... and reduces transmission time // by example • Disk formatter • ... to prepare a disk for use // set up the file system • ... to partition the disc • ... to delete all the data from the disc	3

Answer for Question 25 (9618_w23_12_8)

Question	Answer	Marks
8(c)(i)	1 mark for each bullet point (max 1) • USB / Universal Serial Bus • HDMI	1
8(c)(ii)	1 mark for each component (max 3) Buffers • A buffer temporarily holds data until it is ready to be transmitted to the device Address Bus • The address of the data to be written to the device (in RAM) is carried on the address bus Data Bus • All data to be written to the device / buffer is carried on the data bus	3

Answer for Question 26 (9618_w23_13_7)

Question	Answer	Marks
7(a)	1 mark for each bullet point (max 2). • Static RAM has faster access time • ...because it does not need to be refreshed • ...used on the CPU for improvement of CPU cache speed • Static RAM has lower	2
7(b)	1 mark for each bullet point (max 2) • EEPROM allows frequent / multiple read / write / erase operations • ...which means that the contents of the firmware in the headset can be changed easily • ...without fully erasing the contents of the firmware in the headset first	2
7(c)	1 mark for each bullet point (max 4) • The print instructions and data are sent by the laptop to a buffer (at laptop speed) • The data is transferred from the buffer to the printer (at printer speed) • ...allowing user to continue using the laptop // and allowing processor to continue processing • ...instead of waiting for relatively slower printer • When the buffer is empty an interrupt is sent to the laptop • ...requesting more data	4

Answer for Question 26 (9618_w23_13_7)

Question	Answer	Marks
7(d)	1 mark for naming a correct port 2 marks for matching justification • USB • ...has fast data transfer speeds for data (to the headset) • ...is a universal/popular cable // universal standard • HDMI • ...allows video and audio to be transferred (on the same cable) • ...convenience of HDMI as no need for two cables	3

Answer for Question 27 (9618_s22_11_2)

Question	Answer	Marks
2(a)(i)	1 mark for each correct term. Random Access Memory (RAM) and Read Only Memory (ROM) are both examples of primary memory. One item that is stored in RAM is currently running software/data/part of OS. One item that is stored in ROM is the start-up/boot-up instructions/BIOS. RAM can be either Static RAM (SRAM) or Dynamic RAM (DRAM). SRAM uses transistors arranged as flip-flops/latches. DRAM uses transistors and capacitors.	5
2(a)(ii)	1 mark per bullet point to max 3 - PROM can be set once, EPROM and EEPROM can be overwritten multiple times. - EPROM needs to be removed from device EEPROM can be erased in situ. - EPROM and can be erased using UV light, EEPROM can be erased using voltage // is flash storage . - EPROM must be entirely erased before rewriting, EEPROM does not have to be entirely erased before rewriting.	3

Answer for Question 27 (9618_s22_11_2)

Question	Answer	Marks															
2(b)	1 mark per bullet point to max 5 • The hard disk has (one or more) platter/plate/disk • Each surface of the platter/disk is (ferrous oxide which is) capable of being magnetised • The platters/disks are mounted on a (central) spindle • The entire mechanism is contained inside a sealed (aluminium) box. • The disks are rotated (at high-speed) • (Each surface of the disk) has a read/write head mounted on an arm (positioned just above the surface) • Electronic circuits control the movement of the arm (and hence the heads) • The surface of the platter/disk is divided into <u>concentric</u> tracks / circles • The surface of the platter/disk is divided into sectors • One track in one sector is the basic unit of storage called a block • The data is encoded as a magnetic pattern for each block • When writing to disk, a variation in the current in the head produces a variation in magnetic field on the disk • When reading from disk, a variation in magnetic field produces a variation in current through the head	5															
2(c)(i)	1 mark for each correct row. <table border="1"> <thead> <tr> <th>Gate</th><th>Input 1</th><th>Input 2</th></tr> </thead> <tbody> <tr> <td>AND</td><td>1</td><td>1</td></tr> <tr> <td>NAND</td><td>0 0 1</td><td>0 1 0</td></tr> <tr> <td>XOR</td><td>0 1</td><td>1 0</td></tr> <tr> <td>NOR</td><td>0</td><td>0</td></tr> </tbody> </table>	Gate	Input 1	Input 2	AND	1	1	NAND	0 0 1	0 1 0	XOR	0 1	1 0	NOR	0	0	4
Gate	Input 1	Input 2															
AND	1	1															
NAND	0 0 1	0 1 0															
XOR	0 1	1 0															
NOR	0	0															
2(c)(ii)	1 mark per bullet point • A NAND B • B XOR C • OR (A NAND B) OR (B XOR C)	3															

Answer for Question 28 (9618_s22_12_2)

Question	Answer	Marks												
2(a)	1 mark per point to max 2 - The lane detection system is built into / integrated into the car - The lane detection system only performs one task - The lane detection system is not easily changed/updated by the car owner	2												
2(b)	1 mark for primary - e.g. Miles travelled in the current journey, before the engine is turned off 1 mark for secondary - e.g. Total miles travelled since the car was built // miles for most recent journey after engine switched off	2												
2(c)	1 mark for all correct ticks <table border="1"> <thead> <tr> <th>Statement</th><th>True</th><th>False</th></tr> </thead> <tbody> <tr> <td>The screen always has five different layers</td><td></td><td>✓</td></tr> <tr> <td>A processor determines the horizontal and vertical coordinates of the point of contact</td><td>✓</td><td></td></tr> <tr> <td>The touchscreen will work if any object touches the screen</td><td>✓</td><td></td></tr> </tbody> </table>	Statement	True	False	The screen always has five different layers		✓	A processor determines the horizontal and vertical coordinates of the point of contact	✓		The touchscreen will work if any object touches the screen	✓		1
Statement	True	False												
The screen always has five different layers		✓												
A processor determines the horizontal and vertical coordinates of the point of contact	✓													
The touchscreen will work if any object touches the screen	✓													

Answer for Question 29 (9618_s22_13_2)

Question	Answer	Marks								
2(a)	1 mark for each correct description <table><tr><th>Step</th><th>Description</th></tr><tr><td>$PC \leftarrow [PC] + 1$</td><td>Address in PC is incremented</td></tr><tr><td>$MDR \leftarrow [[MAR]]$</td><td>The data in the address held in the MAR is copied to the MDR</td></tr><tr><td>$MAR \leftarrow [PC]$</td><td>The contents of the PC are copied to the MAR</td></tr></table>	Step	Description	$PC \leftarrow [PC] + 1$	Address in PC is incremented	$MDR \leftarrow [[MAR]]$	The data in the address held in the MAR is copied to the MDR	$MAR \leftarrow [PC]$	The contents of the PC are copied to the MAR	3
Step	Description									
$PC \leftarrow [PC] + 1$	Address in PC is incremented									
$MDR \leftarrow [[MAR]]$	The data in the address held in the MAR is copied to the MDR									
$MAR \leftarrow [PC]$	The contents of the PC are copied to the MAR									
2(b)	1 mark per point to max 5 • Check for interrupt at start/end of an F-E cycle • Priority is checked • If lower priority than current process continue with F-E cycle • If higher priority than current process ... • ... state of current process is / registers are stored on stack • Location/type of interrupt identified • Appropriate ISR is called to handle the interrupt • When ISR finished, check for further interrupts (of higher priority) / return to step 1 • Otherwise load data from stack and continue with next F-E cycle (of process)	5								

Answer for Question 30 (9618_w22_11_7)

Question	Answer	Marks
7(a)	1 mark for each benefit (max 2): • (main) memory requirements for program are reduced as dynamic link library is loaded only once / when required • the executable file size is smaller because the executable does not contain all the library routines • maintenance not needed to be done by the programmer because the DLL is separate from program • no need to recompile the main program when changes are made to DLL because changes / improvements/ error correction to the DLL file code are done independently of the main program	2
7(b)	1 mark for each bullet point (max 2): • RAM is assigned into blocks • dynamic allocation of RAM to programs / processes • reclaims unused blocks of RAM • prevents two programs / processes occupying the same area of RAM at the same time • moves data from secondary storage when needed // manages paging, segmentation and virtual memory	2

Answer for Question 30 (9618_w22_11_7)

Question	Answer	Marks
7(c)	1 mark for each bullet point (max 3): • lossless compression • Run Length Encoding • repeated sequences of <u>characters</u> are replaced by • ... a single copy of the character • ... and a counter of the number of characters	3
7(d)	1 mark for each bullet point (max 2): • cache is fast access memory (close to the CPU) • cache stores frequently used instructions / data • ... more cache means more instructions / data can be transferred faster • ... less swapping between RAM and cache • prevents the CPU idling while waiting for data	2
7(e)	1 mark for each device. • 3D printer: USB port / COM port • Monitor: HDMI / VGA / USB / DisplayPort	2

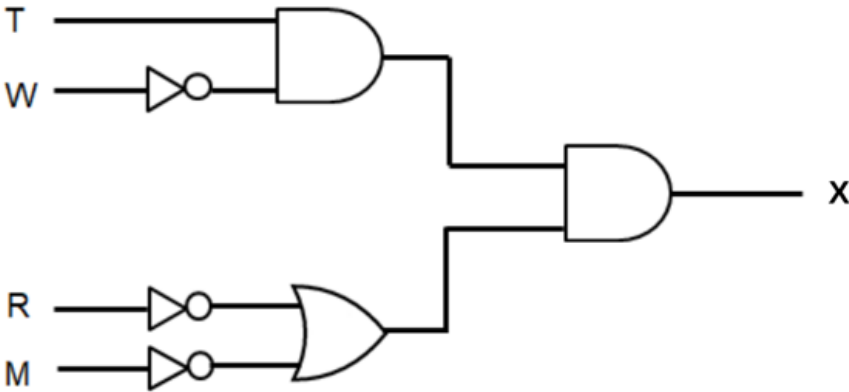
Answer for Question 31 (9618_w22_11_9)

Question	Answer	Marks
9(a)	1 mark for each bullet point (max 2): • the embedded system is built into / integrated into the TV • combination of hardware and software designed for a specific function • The system is not easily changed/updated by the TV owner	2

Answer for Question 31 (9618_w22_11_9)

Question	Answer	Marks
9(b)	1 mark for each benefit and 1 mark for corresponding expansion (max 2). For example: • no additional equipment is needed to change • ... enables firmware updates by non-technical users • can be erased and reprogrammed several times • ... so firmware can be updated • ... can erase a particular byte or the whole EEPROM • possible to reprogram / update • ... without removing it from the device	2

Answer for Question 32 (9618_w22_12_3)

Question	Answer	Marks																																				
3(a)	1 mark for T AND NOT W 1 mark for NOT R OR NOT M 1 mark for final AND 	3																																				
3(b)	1 mark for each set of rows as highlighted: <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	1	0	1	0	1	1	0	0	1	1	1	0	2
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1	1	1	0																																			
3(c)	1 mark for each bullet point: - to store data that does not change - data must be stored even when device is without power - to store boot up instructions / system software / firmware / BIOS	2																																				

Answer for Question 33 (9618_w22_13_10)

Question	Answer	Marks
10(a)	1 mark for each bullet point: • to ensure the system operates with the given criteria • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be <u>automatically</u> adjusted	3
10(b)(i)	1 mark for identification of a suitable sensor 1 mark for corresponding justification Example: • sound sensor • if a sound occurs inside the car the alarm is activated • infra-red sensor • senses the heat of person in the car / infra-red beams are broken • pressure sensor • an intruder sits in the driver's seat	2
10(b)(ii)	1 mark for each bullet point (max 3): • the embedded system is built into / integrated (into the car alarm) • combination of hardware and software designed for a specific function • must have a processor, memory and input / output • The system is not easily changed/updated by the car owner	3

Answer for Question 34 (9618_s21_11_5)

Question	Answer	Marks
5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
5(b)	1 mark for RAM, 1 mark for ROM RAM: Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: Store the start-up instructions (for the washing cycles)	2
5(c)	1 mark per bullet point - The system uses feedback - The system causes the temperature to change // produces an action	2

Answer for Question 35 (9618_s21_13_5)

Question	Answer	Marks
5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
5(b)	1 mark for RAM, 1 mark for ROM RAM: Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: Store the start-up instructions (for the washing cycles)	2
5(c)	1 mark per bullet point - The system uses feedback - The system causes the temperature to change // produces an action	2